

MANVIK TALWAR

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TECHNICAL SKILLS

AI/ML LLMs, Retrieval-Augmented Generation, Agentic Systems, LangGraph, LangChain, Function Calling, Tool Use, MCP, Prompt Engineering, Embeddings, Vector Search, Model Evaluation, PyTorch, TensorFlow, Keras, Hugging Face, OpenCV, MediaPipe, CNNs
Inference llama.cpp, Ollama, GBNF grammar-constrained decoding, Azure OpenAI, Groq, faster-whisper, Quantisation
Backend & Data Python, FastAPI, Node.js, REST APIs, PostgreSQL, MySQL, MongoDB, Redis, LanceDB, ChromaDB, FAISS, BM25
Infra & Frontend Docker, CI/CD, Linux, Azure App Service, Azure Cognitive Search, CUDA, React, Next.js, TypeScript, Tailwind CSS, Tauri

PROFESSIONAL EXPERIENCE

AI Engineering Intern | Niva Bupa Health Insurance May 2026 – Jul 2026

- Built a seven-gate health insurance claims adjudication pipeline running local LLM inference via llama.cpp with GBNF grammar constraints, forcing schema-valid structured output and eliminating free-form parsing failures.
- Designed the gate sequence (Coverage, Financial, Network, Compliance, State Machine) as a hybrid of deterministic rules and semantic checks, keeping auditable decisions in deterministic code and scoping the LLM to interpretation only.
- Refactored financial computation to resolve policy limits from schema at runtime instead of hardcoded constants, and routed low-confidence claims to an assisted-review queue rather than forcing an automated verdict.

Backend Engineering Intern | TechLearn Feb 2026 – May 2026

- Shipped the Dashboard Stats API serving real-time KPIs to educators, now the primary analytics endpoint in production.
- Replaced offset pagination with cursor-based pagination on submission retrieval, removing the deep-page scan cost that grew with dataset size; added composite indexes and restructured trend-analytics queries to cut redundant aggregation passes under load.

AI Development Intern | Path Infotech Jun 2025 – Aug 2025

- Deployed an enterprise RAG assistant to company servers for internal knowledge retrieval, building the semantic search pipeline with custom Azure Cognitive Search indexing skillsets for PDFs.
- Built a citation layer backed by MySQL tagging so every generated answer rendered clickable source references, making outputs auditable for non-technical staff.

SELECTED PROJECTS

DWT-CNN Invisible Image Watermarking | TensorFlow, Keras github.com/CodexManvik/Deep-Learning-Based-Watermarking

- First-authored an end-to-end differentiable encode-attack-decode pipeline embedding a 256-bit payload into the Haar DWT LL sub-band via learned transposed-convolution holographic upsampling; 4.1M parameters, ~18 ms inference per image on an RTX 3050.
- Implemented a six-type differentiable attack simulator including JPEG via straight-through DCT quantisation estimation, so recovery gradients propagate through the attack layer and train the encoder for attack-specific robustness.
- Trained on a 50,000-image mixed-domain corpus (COCO 2017, NIH Chest X-Ray, SIIM CT); reached 37.3 dB PSNR, 0.97 SSIM and 3.2% clean BER, with 2.4x-7x lower BER than a classical DWT baseline across JPEG, noise and rotation. Manuscript in preparation.

Aethel - Local-First AI Agent Runtime | FastAPI, Tauri, llama.cpp github.com/CodexManvik/Aethel

- Built a desktop agent running a fully local inference stack (llama.cpp, faster-whisper, Kokoro TTS, LanceDB) with a step-budgeted tool loop over filesystem, shell and web, governed by a permission manifest and a transaction log with one-click rollback.
- Designed Reflective Skill Memory: the agent reflects on completed tasks into human-reviewable markdown skills, gated by user approval and promoted or retired on tracked outcomes, adapting behaviour without fine-tuning. Validated with an A/B harness scored by programmatic checkers rather than model judging.

Agentic RAG for Distributed Content github.com/CodexManvik/Agentic-Rag-for-Distributed-Content

- Built a multi-agent LangGraph pipeline (normalise, plan, retrieve, adequacy check, reformulate or synthesise, citation validation) over hybrid LanceDB and BM25 retrieval, abstaining explicitly when retrieved context is inadequate.
- Wrote an evaluation harness over a 40-question dataset bucketed by query type, measuring Hit@k, MRR, citation precision and abstain precision/recall; the tuned profile reached 0.690 Hit@k and 0.549 MRR at sub-30s median latency with a 1.000 adversarial block rate.

Interview Mirror - Real-Time Multimodal Coach github.com/CodexManvik/Interview-Mirror

- Fused MediaPipe pose, face-mesh and hand landmarks with faster-whisper speech-to-text and Gemini-generated questions, streaming live posture and delivery feedback over WebSockets to a JWT-authenticated SQLite backend.

EDUCATION

B.Tech, Computer Science (Artificial Intelligence & Machine Learning) | Manipal University Jaipur Aug 2023 – Aug 2027

ACHIEVEMENTS

- **Smart India Hackathon 2025 National Finalist** - FloatChat-AI, a natural-language interface over ARGO oceanographic float data combining semantic search with NL-to-SQL retrieval.
- **4x Student Excellence Award**, Manipal University Jaipur | **Dell x Google Developer Group Ideathon Finalist** | **NPTEL DAA Topper**.